

```
•[5]: nums = list(map(int, input("Enter numbers separated by space: ").split()))

even_nums = [x for x in nums if x % 2 == 0]

odd_nums = [x for x in nums if x % 2 != 0]

largest = max(nums)
smallest = min(nums)

unique_nums = list(set(nums))

print("Even numbers:", even_nums)
print("Odd numbers:", odd_nums)
print("Largest number:", largest)
print("Smallest number:", smallest)
print("List after removing duplicates:", unique_nums)
```

Enter numbers separated by space: 2 3

Even numbers: [2]

Odd numbers: [3]

Largest number: 3

Smallest number: 2

List after removing duplicates: [2, 3]

•[4]:

```
nums = [10, 20, 30]
```

```
nums.append(40)
```

```
print("After append:", nums)
```

```
nums.insert(1, 15)
```

```
print("After insert:", nums)
```

```
nums.remove(20)
```

```
print("After remove:", nums)
```

```
nums.pop()
```

```
print("After pop:", nums)
```

```
nums.reverse()
```

```
print("After reverse:", nums)
```

After append: [10, 20, 30, 40]

After insert: [10, 15, 20, 30, 40]

After remove: [10, 15, 30, 40]

After pop: [10, 15, 30]

After reverse: [30, 15, 10]

```
•[3]: nums = [10, 5, 20, 8, 15]

max_val = max(nums)
print("Maximum value:", max_val)

min_val = min(nums)
print("Minimum value:", min_val)

total = sum(nums)
print("Sum of elements:", total)

asc = sorted(nums)
print("Ascending order:", asc)

desc = sorted(nums, reverse=True)
print("Descending order:", desc)

index_8 = nums.index(8)
print("Index of 8:", index_8)
```

```
Maximum value: 20
Minimum value: 5
Sum of elements: 58
Ascending order: [5, 8, 10, 15, 20]
Descending order: [20, 15, 10, 8, 5]
Index of 8: 3
```

•[2]:

```
a = [1, 2, 3]
```

```
b = [4, 5, 6]
```

```
concat = a + b
```

```
print("Concatenation:", concat)
```

```
repeat = a * 3
```

```
print("Repetition:", repeat)
```

```
membership = 2 in a
```

```
print("Is 2 in list a?", membership)
```

```
print("Length of list a:", len(a))
```

```
print("Length of list b:", len(b))
```

```
Concatenation: [1, 2, 3, 4, 5, 6]
```

```
Repetition: [1, 2, 3, 1, 2, 3, 1, 2, 3]
```

```
Is 2 in list a? True
```

```
Length of list a: 3
```

```
Length of list b: 3
```

```
•[1]: my_list = [10, 3.5, "hello", True]
```

```
print("List:", my_list)
```

```
print("Type of list:", type(my_list))
```

```
for item in my_list:
```

```
    print(f"Value: {item}, Type: {type(item)}")
```

```
List: [10, 3.5, 'hello', True]
```

```
Type of list: <class 'list'>
```

```
Value: 10, Type: <class 'int'>
```

```
Value: 3.5, Type: <class 'float'>
```

```
Value: hello, Type: <class 'str'>
```

```
Value: True, Type: <class 'bool'>
```